

Aryan Garg

agarg54@wisc.edu | [aryan-garg.github.io/](https://github.com/Aryan-Garg) | [linkedin.com/in/aryan-garg019](https://www.linkedin.com/in/aryan-garg019) | github.com/Aryan-Garg

TECHNICAL SKILLS

Programming: Python, Rust, C++, CUDA, TypeScript

Frameworks & Tools: PyTorch, OpenCV, Git, Docker, Hugging Face, XET, Google Cloud, Electron

Large Scale Training: DeepSpeed, FSDP, DDP, LoRA, DreamBooth

Computer Vision & Gen AI: Camera ISPs, HDR, Imaging, Computational Photography, Diffusion, Image Editing, Text-to-Image, Text-to-Video, Video-to-Video, 3D Gaussian Splatting.

EXPERIENCE

Research Assistant

Jan. 2025 – Present

WISION Lab, UW-Madison (Advisor: Prof. Mohit Gupta)

Madison, WI

- gQIR (CVPR 26, Highlight): Increased fidelity of SPAD ISPs by 43% over traditional methods by large scale domain adaptation of Stable Diffusion's VAE and LoRA Adversarial Distillation (1 step) of the Latent Denoising U-Net.
- gQIR (CVPR 26, Highlight): Extended Stable Diffusion to a burst processing capable prior via flow-based warping and motion compensation for temporal stability.
- Radiance Fields From Photons (ToG 25 & SIGGRAPH 26): Adapted NeRF & 3D Gaussian Splatting for 3D reconstruction with single photon cameras.

Research Intern

Jun. 2024 – Aug. 2024

Aalto University (Advisor: Prof. Antti Oulasvirta)

Helsinki, Finland

- Integrated text, segmentation and saliency maps into Stable Diffusion via a Dual Control Adapter: Zero-Encoder and Cross-Attention. Reduced training overhead by ~50% compared to a multi-ControlNet setup.

Research Assistant

Jun. 2023 – Apr. 2024

Computational Imaging Lab, IIT Madras (Advisor: Prof. Kaushik Mitra)

Chennai, India

- HDRSplat (BMVC): Designed a real-time 3D Gaussian Splatting pipeline using RAW DNGs to bypass legacy ISPs. Achieved HDR renderings at 60 fps on consumer GPUs.
- IP-FaceDiff (WACV): Lifted a Text-to-Image prior to text-to-video for facial editing with ArcFace Identity Preservation.
- dpLFVR (ICCP): Dual pixels for light field videos using convolution transformers for consumer mobile 3D video experiences.

MITACS Research Intern

Jun. 2022 – Sep. 2022

LVSN (Advisors: Jean-Francois Lalonde, Yannick Hold Geoffroy)

Quebec, Canada

- Finetuned exiting GAN-based image-to-image baselines (Pix2Pix, SPADE) to generate high dynamic range outdoor skies.

CURRENT PROJECT

Adapting Flux 2 for Controllable HDR Synthesis and Editing. Goal is to push the boundaries of T2I models for high-fidelity, semantically controllable high dynamic range (HDR) generation and editing.

SELECTED PROJECTS

gQIR [🏆★7] CVPR 2026 | Stable Diffusion from Photons. Domain adaptation to SPADs and prior lifting to burst input.

HDRSplat [🏆★46] BMVC 2024 | 3D Gaussian Splatting from raw images for real-time 3D HDR reconstructions

IP-FaceDiff [🏆★3] WACV 2025 | 80% less time Text-to-Video facial editing with identity preservation over multiple edits.

PUBLICATIONS

6 first-author and 10+ papers at top-tier vision/AI venues: IEEE/CVF CVPR, ACM SIGGRAPH, ICLR, BMVC, WACV

HONORS & AWARDS

2026 – Accepted into the first FLUX Creator's cohort from Black Forest Labs

2025 – UW-Madison, CS Department Summer Research Fellowship Award

2025 – Qualcomm Innovation Fellowship - North America Finalist (Out of 266 teams)

2023 – Awarded Bosch AI Fellowship (RBCDSAI) at IIT Madras

2019 – JEE Mains Rank: 1834 among 1.1 million students. JEE Advanced Rank: 5137 for IIT selection.

EDUCATION

University of Wisconsin-Madison

Madison, USA

Ph.D. in Computer Sciences

May. 2026 – Current

University of Wisconsin-Madison

Madison, USA

M.S. in Computer Sciences

Sep. 2024 – May. 2026

Indian Institute of Technology, Mandi

Mandi, India

B.Tech. in Computer Sciences and Engineering

Aug. 2019 – May 2023